Assignment 5: Part 1 and 2

A Comparative Analysis of Base R and ggplot2

1. Histogram

Customizing histograms is more straightforward in ggplot2, as geom_histogram() enables direct adjustment of binning, colors, and annotations. In contrast, Base R's hist() function suits basic charts, but customization requires manual effort.

2. Vertical Bar Chart

ggplot2 simplifies creating vertical bar charts, as geom_bar() automatically counts categories and geom_text() adds value labels. In Base R, barplot() requires more manual intervention to position labels and modify the chart's appearance.

3. Horizontal Bar Chart

ggplot2 facilitates creating horizontal bar charts through coord_flip(), which reorients a vertical chart without altering the data. In Base R, setting horiz = TRUE is straightforward, but further adjustments to labels and formatting require additional steps.

4. Pie Chart

ggplot2 provides multiple options for styling pie charts using geom_col(), coord_polar(), and geom_text(), which simplifies customizing labels and legends. Base R's pie() function is more direct for simple charts, but advanced formatting requires additional effort.

5. Boxplot

ggplot2 streamlines customizing and presenting boxplots, as geom_boxplot() efficiently handles grouped data and supports consistent color schemes and themes. While Base R's boxplot() suits basic applications, achieving a comparable appearance requires more manual adjustments.

6. Scatterplot

ggplot2 facilitates enhancing scatterplots, as geom_point() and geom_smooth() add data points and regression lines as distinct layers. Base R can achieve similar results using plot() and abline(), but adding legends and advanced styling requires multiple commands.

Overall Comparison

Base R is convenient for quickly generating simple charts without additional packages. However, ggplot2 generally produces more e effective, polished, consistent, and customizable visualizations, as its layered structure makes it easier to add and manage other graphical elements.

Assignment 5: Part 3

Part 3: Comparison of Export Formats

1. Comparison of File Formats

The scatterplots created using base R and ggplot2 were exported in PDF, SVG, PNG, JPG, TIFF, and BMP formats. For both, charts were exported at the same size, 6.5 × 4 inches, with raster images rendered at 300 dpi to ensure a fair comparison.

The following table gives a summary of each format.

Format	Type	File size (KB)	Compression	Transparency support
PDF	Vector (typically)	Record actual size	Can use lossless compression	Yes
SVG	Vector	Record actual size	Usually uncompressed; can be compressed separately	Yes
PNG	Raster	Record actual size	Lossless	Yes
JPG/JPEG	Raster	Record actual size	Lossy	No
TIFF	Raster	Record actual size	Lossless with LZW compression	Yes, depending on export settings
BMP	Raster	Record actual size	Usually uncompressed	Generally no

PDF and SVG are vector formats, which preserve the sharpness of text and lines when resized. Raster formats store images as pixels, so their quality depends on resolution. Transparency support for PDF and TIFF can depend on the export settings and the software used to display the file.

2. Observations at 400% Zoom

Examining each exported file at 400% zoom highlights the differences between vector and raster graphics.

- **PDF:** Text and lines generally remain sharp and clear because vector elements can be scaled without becoming pixelated. Font substitution may affect the appearance of some text.
- **SVG:** Text and lines remain sharp at high zoom levels because the image is described using scalable vector elements. Its appearance may vary slightly across applications depending on font availability and rendering.
- **PNG:** Text and lines are generally clear at normal viewing sizes. At 400% zoom, individual pixels may become visible, although the lossless compression preserves the original pixel information.
- **JPG/JPEG:** Text edges and thin lines may appear blurred or show compression artifacts, especially around high-contrast boundaries. This occurs because JPEG uses lossy compression.

- **TIFF:** At 300 dpi, text and lines generally appear clear, but pixelation may become visible at 400% zoom. LZW compression preserves the original raster information without loss.
- **BMP:** Text and lines may appear pixelated at 400% zoom because the image is raster-based. BMP files are often larger than compressed raster formats because they are usually stored without compression.

Overall, PDF and SVG provide better results for zooming and resizing, while PNG and TIFF are appropriate for high-quality raster output. JPEG is less suitable for plots with small text and thin lines because compression reduces clarity.

3. Recommended Formats for Different Destinations

Journal article — PDF or TIFF

PDF is recommended when the journal accepts vector graphics because it preserves sharp text, lines, and symbols at various sizes. TIFF is appropriate when a journal requires high-resolution raster images, such as 300 dpi or higher. The journal's submission guidelines should determine the final format.

Web page — SVG or PNG

SVG is better for charts on web pages because it scales well across screen sizes and keeps text and lines sharp. PNG is a suitable alternative when broad compatibility and consistent appearance are priorities. PNG also supports transparent backgrounds when exported with transparency enabled.

PowerPoint slide — PNG or SVG

PNG is a reliable option for PowerPoint because it is widely supported and provides high image quality at 300 dpi. SVG is preferable when the presentation software supports it and the chart may require resizing, as it retains sharpness. PDF can also be used in some workflows, but PNG or SVG is generally more convenient for inserting charts directly into slides.

Conclusion

This comparison demonstrates that the optimal export format depends on the intended application. Vector formats such as PDF and SVG are preferable when scalability and sharp text are required. Raster formats such as PNG and TIFF are better for fixed-resolution images. JPEG is appropriate for photographs but less suitable for charts with fine lines and labels, while BMP is less efficient due to larger file size.